

SIH26145: AI-Based Detection of Cyber Threats in Unidirectional IP Traffic

Objective: Build an AI-based system that monitors unidirectional IP traffic, extracts network-flow features, detects abnormal behavior, classifies potential cyber threats such as DoS/DDoS and Port Scanning, assigns a risk/confidence score, and displays the results through a real-time dashboard.

Technical Terms In Problem Statement:

What is IP?

IP usually means Internet Protocol.

An IP address is a unique address given to a device on a network so that data knows where to go and where it came from.

What is IP Traffic:

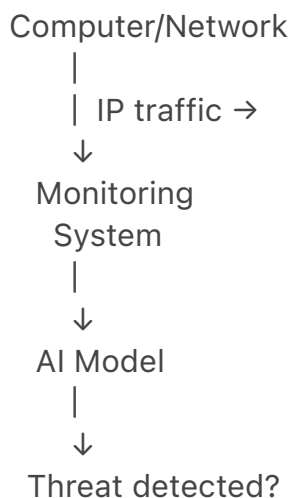
IP traffic means the data packets being sent and received using the Internet Protocol (IP) across a network.

What Is Unidirectional IP traffic:

Unidirectional IP traffic means IP traffic that primarily flows in one direction between two endpoints.

Actual Problem:

Imagine:



The system receives IP traffic, particularly traffic that is unidirectional—traffic primarily flowing in one direction.

The challenge is:

Can we automatically detect whether that traffic contains signs of a cyberattack?

Instead of relying completely on manually written security rules, the idea is to use AI/ML to identify suspicious patterns.

What does unidirectional IP traffic contain?

1. Packet rate -How many Data packets are being sent every second
2. Packet size - How much data is inside one packet
3. Destination IP - The IP address where the packet is going
4. Destination port - A port tells the computer which services/Application Should receive the data.(IP- Building, Port- Room No.)
5. Protocol - The rules/method used for communication
6. Bytes transferred - Total amount of data that was sent
7. Flow duration - How long a particular communication lasts
8. Source Ip- from where traffic is coming from
9. Packets per flow- how many packets are exchanged
10. Inter-arrival time- how Much time between each packet
11. Unique destination ports- The number of different port numbers a device is trying to communicate with on the destination
12. Network Flow- A flow is a group of packets belonging to the same communication

How it will Work:

Network Traffic



Feature Extraction



Source IP

Destination IP

Source port

Destination port

Protocol

Packet count

Packet rate

Average packet size

Total bytes transferred

Flow duration

Inter-arrival time

Unique destination ports



ML Model



Normal?

Suspicious?

Malicious?

Threats to be focused:

- DDoS / DoS- Huge amounts of traffic:
- Data exfiltration- A compromised machine may send unusually large amounts of data outward
- Malware communication - A compromised machine might repeatedly communicate with suspicious destinations
- Scanning / reconnaissance - A machine may communicate with many ports/IPs To gather Info

Threats To be Focused In Prototype:

1. DDoS / DoS
2. Port Scanning

Technologies Needed For Prototype:

Component	Technology
Frontend	React
Styling	Tailwind CSS
Charts	Recharts
Backend	Python + FastAPI
API	REST API
Database	SQLite
Database ORM	SQLAlchemy
ML	Scikit-learn
ML Algorithm	Random Forest
Data Processing	Pandas + NumPy
Traffic Capture	PyShark / Scapy
Packet Analysis	Wireshark / Tshark
Model Storage	Joblib
Communication	JSON

How it will work:

